DistilBERTClassifer on 4-way News Topic Classification

Model Details

- **Model Name:** DistilBERTClassifer.
- **Base Model:** `distilbert-base-uncased` from HuggingFace.
- **Developers:** Teun Boersma (s5195179), Julian Sprietsma (s5096219), Marcus Harald Olof Persson (s5343798).
- **Model Type:** Transformer-based sequence classifier.
- **Task:** Multi-class text classification (4 classes).
- **Language:** English.

Intended Usage and Limitations

This model's primary usage is for the classification of news articles into four specific classes: World, Sports, Business, and Sci/Tech. Regarding scope, the model is specifically finetuned on this dataset and may not generalise to non-news domains or different language contexts without further finetuning.

Training Data

The dataset originates from the AGNews dataset. For preprocessing, the model utilises a combined "text" field created by concatenating the "title" and "description" columns from the dataset. With labelling of data:

- 1: World.
- 2: Sports.
- 3: Business.
- 4: Sci/Tech.

Training Procedure

The model was trained using a custom `Trainer` class on the University of Groningen's high-performance computing cluster Hábrók.

- **Hardware:** 1x NVIDIA A100 GPU, 4 CPUs, 16GB RAM.
- **Optimiser:** AdamW with a learning rate of 2×10^{-5}.
- **Loss Function:** `CrossEntropyLoss` .
- **Batch Size:** 32.
- **Early Stopping:** Triggered after 4 epochs without improvements, where the best model weights being restored from epoch 3.

Epoch	Train Loss	Eval/Valid Loss	Note
1	0.2352	0.2636	
2	0.1410	0.2690	
3	0.1011	0.2356	Best Model
4	0.0695	0.2955	
5	0.0478	0.3328	
6	0.0331	0.3674	Early stopping triggered

Results

	Dev Set	Test Set
Accuracy	0.9226	0.9424
F1 Score (Weighted)	0.9224	0.9424

Predicted \ Actual	World	Sports	Business	Sci/Tech
World	1785	14	57	44
Sports	10	1880	6	4
Business	32	8	1724	136
Sci/Tech	26	8	93	1773

Biases and Risks

- The model inherits potential biases from the `distilbert-base-uncased` pretraining data.
- Performance may drop significantly if applied to news data significantly newer than the training set.
- The model most frequently confuses "Business" news with "Sci/Tech".

Environmental Impact

- **Data Centre:** Hábrók is maintained at the Coenraad Bron Center, which is reported to operate at a Power Usage Effective (PUE) of 1.25 or below.
- **Location:** Groningen, the Netherlands.
- **Total Runtime:** Approximately 51 minutes and 52 seconds, rounded up to an hour.
- **Carbon Emitted:** From the Machine Learning Emissions Calculator, assuming the carbon effiency is OECD's 2014 yearly average ($0.432 \text{ kg}/\text{kWh}$), then $0.11 \times 1.25 = 0.1375 \text{ kg CO}_2$ was approximately emitted.